# SHIVANESH G

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## Education

- **REVA University** — Bangalore, India
  Bachelor of Technology - Artificial Intelligence and Data Science; CGPA: 9/10 — 2023 - Present
- **Sri Chaitanya Techno School** — Bangalore, India
  Class 12th - CBSE; Grade: 85% — 2023
- **Infant Eng Med H S** — Bangalore, India
  Class 10th - KSEEB; Grade: 93% — 2021

## Skills Summary

- **Languages & Core CS**: Python, Java, C, JavaScript, SQL, Data Structures & Algorithms (250+ Problems Solved), Object-Oriented Programming, DBMS, Operating Systems, Software Design
- **AI/ML & Data Engineering**: Machine Learning, Deep Learning, Computer Vision, NLP, PyTorch, Generative AI, Apache Spark, ETL Pipelines, Time-Series Analytics, Predictive Maintenance, Hyperparameter Optimization, Feature Engineering
- **Cloud & Infrastructure**: GCP(Vertex AI, BigQuery, Cloud Run, Cloud Storage, Pub-Sub), Distributed Systems, AWS
- **Frameworks & Tools**: TensorFlow, React, Firebase, Next.js, Node.js, PostgreSQL, Git, GitHub, REST APIs, Postman

## Experience

- **Gas Turbine Research Establishment - DRDO** — Bangalore, India
  AI/ML Intern (Engine Data Systems & Diagnostics) — July 2025 - August 2025
  - **Predictive Maintenance & RUL Modeling**: Tackled an open-ended modeling problem, developing and benchmarking Linear Regression, Random Forest, XGBoost, LSTM, and FCNN models with hyperparameter optimization on NASA C-MAPSS datasets for Remaining Useful Life prediction, achieving a best RMSE of 12.87 and $R^2$ score of 0.83.
  - **Condition Monitoring & Diagnostics**: Analyzed high-dimensional sensor streams from gas turbine systems to detect performance degradation, validate sensor behavior, and support engine health assessment workflows.
- **Google Developer Group on Campus REVA University** — Bangalore, India
  Lead (Organizer) — Sept 2025 - Present
  - **Technical Leadership, Stakeholder Engagement & Mentorship** : Led 25+ cloud and AI learning initiatives, mentored 400+ students on Google Cloud technologies, organized workshops and hackathons, and drove adoption of cloud-native development practices.
  - **Strategic Partnerships & Events** : Collaborated with Google for Developers, GDG Bangalore, Rotaract Club of Bangalore, and Hack2skill to deliver 3+ hackathons, bootcamps, and developer-focused initiatives.

## Projects

- **Scarlet Link (Emergency Blood Network)** [**Link**][**Github**]: (International Hackathon Winner) Architected a distributed, fault-tolerant matching system ranking 30,000+ donors via LightGBM scoring on compatibility, ETA, and response probability; achieved sub-2-second response via Pub/Sub, Firebase Cloud Messaging, and Redis-cached inventory on Cloud Run.
- **Cloud Jams Progress Tracker** [**Link**][**Github**]: Production platform used by 300+ students to track Google Cloud Skills Boost progress; designed indexed PostgreSQL schema, implemented rate limiting for concurrent API requests, incremental Redis caching with scheduled background refresh, and automated badge scraping with CSV ingestion, deployed on Vercel with SSR.
- **Scarlet Save (AI Powered Anemia Diagnostics)** [**Github**]: Non-invasive anemia screening system trained on 12,000+ images using Vertex AI AutoML Vision; classifies anemia severity with 88% accuracy and supports recovery tracking.
- **AI-Based Flood Region Detection System** [**Github**]: Built a U-Net model on Sentinel-1 SAR imagery, achieving 91% segmentation accuracy and 0.87 IoU for automated flood detection, mapping, and disaster monitoring.

## Publications

- **Research Paper: Hybrid LLM-ML Framework for Remaining Useful Life Prediction of Turbofan Engines**: Published at NCCM 2025; proposed a hybrid LLM-ML framework leveraging NASA C-MAPSS datasets for Remaining Useful Life prediction, condition monitoring and maintenance decision support in turbofan engines with RMSE of 12.87 on test set

## Achievements

- **Best Paper Award – NCCM 2025**: GTRE-DRDO, Condition Monitoring Society of India
- **International Hackathon Winner – AI For Good 2025**: Winner for Scarlet Link
- **National Talent Award**: Karnataka Prathibha Academy

## Certifications

- **Google Specialization**: Advanced Data Analytics, IT Automation with Python, Project Management, Business Intelligence
- **Google Cloud Skills Boost**: GenAI Apps, Vertex AI, Cloud Run, Pub/Sub, API Gateway, Cloud Storage

## Volunteer Experience

- **Build For Bengaluru 2026 & Build with AI 2026 Bootcamp**: Organized large scale hackathons and AI bootcamps.
- **Google Cloud Facilitator, Mentor & Judge**: Mentored 200+ students in GCP and served as judge for multiple hackathons